

Sprint Resisted and Assisted Priming for Peak Performance

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Abstract

Krzysztof, K, Aleksander, M, Adam, Z, and Krzysztofik, M. Sprint resisted and assisted priming for peak performance. *J Strength Cond Res* 37(12): 2354–2361, 2023—This study aimed to investigate resisted and assisted sprint, or a combination of both, as a conditioning activity (CA) on creatine kinase activity, biceps femoris stiffness, postactivation performance enhancement (PAPE), and priming in a sprint. Ten female sprinters (age: 20.1 ± 2.3 years, body mass: 56.4 ± 4.2 kg, body height: 171 ± 5 cm, training experience: 5.6 ± 2 years) participated. Before the warm-up, blood samples were drawn to determine creatine kinase at baseline. After the warm-up, biceps femoris stiffness was determined, and consequently, 50-m sprint time (with 0- to 20-m and 20- to 50-m split time evaluation) was assessed 5 minutes before (pre-CA) and in the seventh minute after the completion of each CA (4 sets of 40 m either: 10% body mass resisted [RST]; 105% maximum velocity assisted [AST] sprint; or 2 sets of each sprint as the CA [COMB]), as well as after 48 hours of recovery. Furthermore, another blood sample was taken 48 hours later, before the warm-up, to assess changes in creatine kinase activity. A significantly higher 20-m sprint time at post-CA ($p = 0.006$; ES = 0.4) and post-48 ($p = 0.011$; ES = 0.59) compared with pre-CA in the RST condition was found, whereas a significantly lower sprint time was reported in post-48 compared with post-CA ($p = 0.019$; ES = 0.44) in the COMB condition. However, a significantly lower 30-m flying sprint time at post-CA ($p < 0.001$; ES = 0.28) and at post-48 ($p = 0.014$; ES = 0.22) compared with pre-CA in the RST condition was revealed. Moreover, a significantly lower 30-m flying sprint time at post-48 compared with the pre-CA ($p = 0.007$; ES = 0.7 and $p = 0.003$; ES = 0.77) and with the post-CA ($p = 0.019$; ES = 0.61 and $p = 0.009$; ES = 0.68) in the AST condition and COMB was reported. Furthermore, a significantly lower 50-m sprint time at post-48 compared with pre-CA ($p = 0.015$; ES = 0.51 and $p = 0.011$; ES = 0.64) and with post-CA ($p = 0.005$; ES = 0.46 and $p = 0.001$; ES = 0.55) in the AST and COMB condition were found. The creatine kinase activity at post-48 during AST was significantly lower than during RST ($p = 0.028$; ES = 0.73) and COMB ($p = 0.028$; ES = 1.48). No significant changes were found for biceps femoris stiffness ($p = 0.085$; $\eta^2 = 0.199$). The COMB and AST contributed to the improvement of the 50-m sprint time 48 hours after activation, which may be a good precompetition approach. However, during training sessions aiming to induce an acute PAPE effect and improve the 30-m flying sprint time, the best solution might be to use low-volume-resisted sprints.

Key Words: postactivation performance enhancement, conditioning activity, running, track and field, postactivation potentiation

Introduction

Elite athletes engage in long-term periodized training programs with the objective of developing the most significant motor abilities for a certain sport or a particular event (16). Such is the case in athletics, where sprinters use different modes of resistance training throughout the annual training cycle to develop high rates of force production necessary for acceleration and maximal running velocity (6). To increase performance, coaches use the supercompensation phenomena, which asserts that an athlete who pairs his training load with adequate recovery time will enhance his fitness level and systematically improve their performance. In supercompensation, the athlete can handle the same training load or a greater load with ease in subsequent workouts if recovery is adequate and the new stress is timed properly. This adaptive phenomenon is an ongoing wavelike process that allows the long-term development of fitness and performance (2). However, the training objective of competitive athletes is to reach

peak performance during a particular training session and especially during priority competitions.

Postactivation performance enhancement (PAPE) is a strategy that allows for enhancing performance acutely because of completing a low-volume, high-intensity resistance exercise minutes before performing an explosive activity, such as a sprint (34,36). Postactivation performance enhancement strategies have been used extensively in different sports disciplines, enhancing jumping and sprinting performance (14,38). Most research has been conducted on team sport athletes, whereas PAPE data from elite sprinters are scarce. In athletics and especially in track sprints, PAPE research has concentrated on 2 types of conditioning exercises. They include resistance (such as back squats) (43) or plyometric exercises (i.e., drop jumps, alternate bounds) (5,11) and locomotor resistance exercises, which are usually biomechanically more specific for sprinting. These include bounding, sled towing, uphill sprinting, or harness running (4). Recently, Matusiński et al. (26,27) evaluated the PAPE effects of resisted and assisted sprinting on starting speed, maximum running velocity, and power in elite female and male sprinters using the 1,080 Sprint drag technology for conditioning activities. The results were quite promising, and suggestions for optimizing sprint

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loading and rest intervals between the conditioning activity and the sprint were suggested (26,27). Postactivation performance enhancement seems a great training modality to be applied in training. However, during competition, the small window opportunity to enhance performance makes it difficult or even impossible to be applied. However, there is some theoretical and empirical evidence that another window of potentiation exists for up to 48 hours following a resistance exercise stimulus (10,16).

Recent studies have suggested that numerous neuromuscular performance variables may be enhanced following resistance exercise for up to 48 hours after the conditioning activity (25,31). On the contrary to PAPE, which lasts up to 12 minutes (33), the phenomenon described above enhances performance for up to 48 hours and has been termed priming (39). Previous studies indicated that highly loaded resistance or high-intensity exercise intending to perform a movement at high velocity is effective in eliciting PAPE (23,24) and priming (10,31). For example, Cook et al. (10) compared sprints (3 maximal 40-m sprints) and resistance (3 one-repetition maximum bench press and back squat) morning priming sessions on the afternoon (6-hour later) back squat, bench press, 40-m sprint, and countermovement jump performance in rugby union players. The resistance priming session improved performance in all measures, whereas the sprint session increased only sprint performance. Furthermore, the magnitude of improvement in the 40-m sprint time was greater after resistance priming. By contrast, Russell et al. (31) compared resistance (5 sets of 10 repetitions of the bench press exercise at 75% one-repetition maximum), cycling (6 sets of 6-second cycle ergometer sprints), and sprinting (6 maximal 40-m sprints with a 180° change of direction at the midway point) priming sessions in rugby players. The results showed that sprint priming improved sprint performance and jump height 5 hours later. The differences between these studies may indicate that using free sprints for priming requires a relatively high volume of approximately 6 sets instead of 3 sets, as in the study by Cook et al. (10) or the use of resisted sprints. Moreover, despite the higher volume in the study by Russell, each of the priming sprints was interspersed with a 180° change of direction; thus, subjects had greater force requirements because of multiple decelerations and accelerations. Surprisingly, despite that priming might have significant implications for athletes as they prepare for competition to maximize performance, most studies evaluated relatively short priming window delays (~6-hour) (9,12,14,28,29), which might be impractical in scenarios when the competition starts in the morning. In addition, to the best of the authors' knowledge, there is a lack of data on the effectiveness of sprint-priming sessions among sprinters, although it could be of great interest considering that minimal differences in sprint events are of great importance (29).

Moreover, little data are available on multiple contractions using stretch-shortening cycles as required in sprinting. Theoretically, conditioning activities that require repeated bouts of the stretch and shortening cycles, as in resisted sprinting, may reduce potentiation because of fatigue. Given the meaningful impact of conditioning activity-induced fatigue, using a method that could noninvasively assess its magnitude would be helpful. One such method might be myotonometry, which has been shown to be a simple and reliable method for determining the viscoelastic properties (i.e., stiffness, tone) of particular muscles (3). Through a movable indentation probe, the portable myotonometer device assesses the mechanical response of the tissue to a brief mechanical perturbation. Myotonometric measurements show a near-linear correlation with surface electromyography activity and therefore provide an indirect measure of changes in the ability of

muscles to generate force (3,20). Previous research has shown that fatigue may be indicated by increased muscular stiffness and tone (37,40,41). Because of increased tension and contraction, high tone is associated with increased intramuscular pressure. Restricted blood flow because of elevated intramuscular pressure causes early onset of muscle fatigue, slower muscle recovery, and decreased strength (19). Therefore, monitoring changes in muscles engaged during conditioning activity and identifying the time when fatigue has subsided will increase the chances of performance improvement as long as the potentiated state occurs.

Considering that little data are available on resistance priming and sprint performance regarding sprinters, we attempted to compare the acute and delayed (48-hour later) effects of 3 different configurations of 40-m sprints as precompetition activation protocols: (a) 4 sets of 10% body mass resisted; (b) 4 sets of 105% maximum velocity assisted sprint; (c) and 2 sets of assisted and 2 sets of resisted sprints, on acute and delayed (48 hours later) 50-m sprint time with 0- to 20-m and 20- to 50-m split time assessment, biceps femoris stiffness, and creatine kinase concentration. Given that previous studies showed the effectiveness of resisted sprints in inducing the PAPE effect (26,27), we hypothesized that they will also be effective in this study eliciting PAPE and priming effect.

Methods

Experimental Approach to the Problem

The experiment was performed following a randomized cross-over design where resisted, assisted, or a combination of both was used as a conditioning activity. This setup aimed to investigate the effects of a particular priming session on creatine kinase activity, biceps femoris stiffness, and subsequent 50-m sprint time (with 0–20-m and 20- to 50-m split time assessment) among trained female sprinters. The randomization was performed for 3 different conditions using automatic software (www.randomizer.org).

Before the warm-up, blood samples were drawn to determine creatine kinase activity. After the warm-up, biceps femoris stiffness was determined, and consequently, 50-m sprint time (with 0- to 20-m and 20- to 50-m split time evaluation) was assessed 5 minutes before (pre-CA) and in the seventh minute after the completion of each CA (resisted: RST; assisted: AST; combined: COMB), as well as after 48 hours of recovery. Furthermore, another blood sample was taken 48 hours later, before the warm-up, to assess changes in creatine kinase activity (Figure 1).

Subjects

Ten female sprinters (from 100 to 400 m) were selected to participate in this study (age: 20.4 ± 2.3 years, body mass: 56.4 ± 4.2 kg, body height: 171 ± 5 cm, training experience: 5.6 ± 2 years). The inclusion criteria were as follows: (a) free from neuromuscular disorders, (b) belongs to the national team, (c) participation in national and international competitions in the past year, (d) taking part in regular sprint training for the past year, and (e) had previous experience with resisted and assisted sprint training. Athletes were instructed not to perform any resistance exercises 72 hours before testing to avoid fatigue, maintain their usual dietary and sleep habits, and not to use any caffeine-rich drinks or supplements before the assessments. Subjects were allowed to withdraw from the experiment at any time and were informed of the benefits and potential risks of the study before providing their

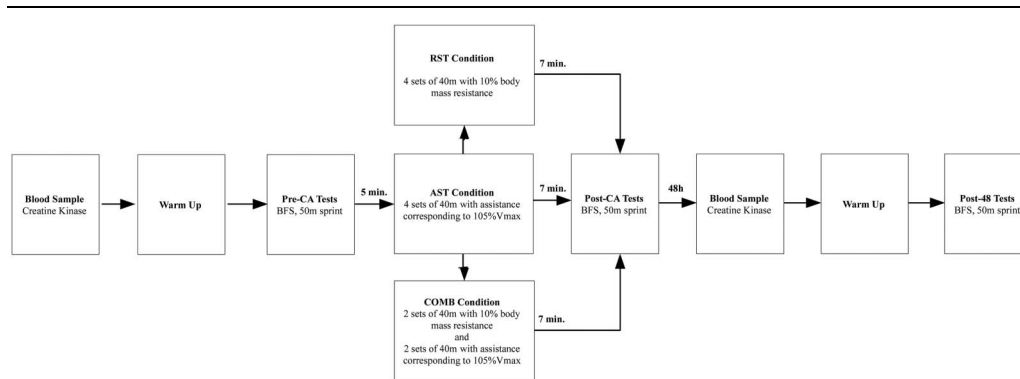


Figure 1. Schematic representation of the study design. CA = conditioning activity; BFS = biceps femoris stiffness; RST = resisted conditioning activity; AST = assisted conditioning activity; COMB = combined resisted and assisted conditioning activity.

written informed consent for participation. Subjects were not informed of the expected study outcomes. The protocol was approved by the Bioethics Committee for Scientific Research at the Jerzy Kukuczka Academy of Physical Education and performed according to the ethical standards of the Declaration of Helsinki 2013.

Experimental Sessions

The subjects performed 3 different testing conditions (1 week apart) in a randomized order on an indoor synthetic track. The same typical pretraining warm-up was performed before each experimental session, which consisted of 5 minutes of jogging followed by 10 minutes of dynamic exercises, including side and forward bends, hip thrusts, lunges, and 30 m A, B, and C skipping drills, 30-m submaximal sprints. After that, the tests were performed in the following order: baseline biceps femoris stiffness and 50-m sprint performance measurements, CA, post-CA evaluation of biceps femoris stiffness, and 50-m sprint performance assessment in the seventh minute after the CA. After 48 hours of recovery, creatine kinase activity, the biceps femoris stiffness, and 50-m sprint performance were retested. During each 50-m sprint, 0- to 20-m and 20- to 50-m split times were also evaluated. The subjects performed 4 sets of 40 m (a) 10% body mass RST, (b) 105% maximum velocity assisted sprint (AST), or (c) 2 sets of assisted and 2 sets of resisted sprints as the CA (COMB). Five-minute rest intervals between each set were allowed. During particular sprints, the 1,080 Sprint engine-assisted measuring system (1,080 Motion AB, Stockholm, Sweden) was used for the precise selection of loads.

Measurement of Biceps Femoris Stiffness

The handheld myometer (MyotonPRO, Myoton AS, Tallinn, Estonia) was used for the noninvasive assessment of the biceps femoris stiffness of the dominant limb through superficial mechanical deformation. The device's probe was placed perpendicular to the skin, and brief mechanical compression was applied (0.4 N for 15 milliseconds), with a constant preload force of 0.18 N. After a mechanical impulse, the soft tissue responds in the form of a dampened oscillation, which is measured by an accelerometer. The measurement for the biceps femoris was at 50% of the distance from the isocenter to the lateral epicondyle of the femur (42). A permanent marker was used to mark the measurement point. Moreover, the subjects were instructed to correct the mark

if it started to blur. Measurements were made in a state of muscle relaxation, with subjects lying in a prone position by the same researcher at every time.

Measurement of Sprint Time

Sprint times were recorded using timing photocells (Microgate, Bolzano, Italy), with gates at 0, 20, and 50 m. The height was set at approximately 1 m off the ground, corresponding to subjects' hip height, to avoid the timing gates being triggered prematurely by a swinging arm or leg. The subjects started from the crouched position 0.3 m behind the first timing gate to prevent any early triggering of the photocells. Times were measured to the nearest 0.01 seconds. Two attempts were performed, and the best 50-m sprint performance was kept for further analysis.

Measurement of Creatine Kinase

Approximately 5 ml of venous blood was sampled from the antecubital fossa vein before the priming session and 48 hours later before the start of the second session. The blood was immediately placed into test tubes containing anticoagulant (EDTA), inverted and rolled, then transferred into Eppendorf tubes and centrifuged at 3,400 rpm for 20 minutes. The activity of creatine kinase was measured on the Alinity c analytical system (Abbott Laboratories, Abbott Park, IL).

Statistical Analysis

A Shapiro-Wilk test was used to test the normal distribution of the data, and Mauchly's test was used to test for the assumption of sphericity. A single-rater intraclass correlation coefficient (ICC) and coefficient of variability (CV) were used to measure value reliability (calculated from the baseline measurements taken for each of the dependent variables) (19). The thresholds for interpreting ICC results were <0.5 "poor," 0.5–0.75 "moderate," <0.76–0.9 "good," and >0.90 as "excellent" (19). For coefficient of variation, results were <10% "very good," 10–20% "good," <21–30% "acceptable," and >30% "not acceptable" (7).

Two-way analysis of variances (ANOVAs) ($3 \times$ conditions [RST; AST; COMB] $\times$ 3 time points [pre-CA; post-CA; post-48]) were used to investigate the influence of CA on biceps femoris stiffness and sprint time. Because the normality was not

confirmed for creatine kinase, related-samples Friedman's 2-way ANOVA by ranks was used; ESs were estimated by Kendall's coefficient of concordance. When a significant main effect or interaction was found, post hoc tests with Bonferroni's correction were used to analyze the pairwise comparisons. The magnitude of mean differences was expressed with standardized effect size by use of the following formula:

$$g = \frac{M_1 - M_2}{SD^*_{\text{pooled}}} \times \left(\frac{N-3}{N-2.25} \right) \times \sqrt{\frac{N-2}{N}},$$

where M is mean; SD^*_{pooled} is pooled and weighted standard deviation; N is the number of subjects. Thresholds for qualitative descriptors of Hedges g were interpreted as ≤ 0.20 "small", 0.21 – 0.79 "medium", and > 0.80 as "large" (9).

All data were analyzed using IBM SPSS Statistics for Macintosh, Version 25.0 (IBM Corp., Armonk, NY) and are reported as the means with SD s and their 95% confidence intervals (CIs). Statistical significance was set at $p < 0.05$.

Results

The sample size was determined using G*Power version 3.1.9.2 (Dusseldorf, Germany), and the parameters, such as "ANOVA, repeated measures, within factors," were assumed as a statistical test (1 group of subjects, 3 experimental conditions, and 3 measurements); the statistical power was 0.8, the significance level was 0.05, and the effect size (ES) was 0.51 based on the PAPE effect on sprinting performance (34). The power analysis indicated that a minimum sample size of 9 individuals was required for this study. The measurement reliability was good to excellent for sprint time (ICC = 0.81 – 0.95, CV = 1.3 – 1.9%), biceps femoris muscle stiffness (ICC = 0.95, CV = 3.8%), and creatine kinase activity (ICC = 0.92, CV = 7%).

Sprint Time

The 2-way ANOVA indicated that there was a statistically significant interaction of 20-m sprint time ($F = 4.34$; $p = 0.006$; $\eta^2 = 0.325$). The post hoc analysis revealed a significantly higher 20-m sprint time at post-CA ($p = 0.006$; ES = 0.4) and

post-48 ($p = 0.011$; ES = 0.59) compared with pre-CA in the RST condition. Moreover, the 20-m sprint time was significantly lower in post-48 compared with post-CA ($p = 0.019$; ES = 0.44) in the COMB condition. Furthermore, a significantly lower 20-m sprint time at post-CA ($p = 0.026$; ES = 0.36) and post-48 ($p = 0.006$; ES = 1.04) was noted in the COMB condition compared with the RST condition corresponding time points (Figure 2).

The 2-way ANOVA indicated that there was a statistically significant interaction of 30-m flying start sprint time ($F = 3.893$; $p = 0.01$; $\eta^2 = 0.302$). The post hoc analysis revealed a significantly lower 30-m flying sprint time at post-CA ($p < 0.001$; ES = 0.28) and post-48 ($p = 0.014$; ES = 0.22) compared with pre-CA in the RST condition. Moreover, a significantly lower 30-m flying sprint time at post-48 compared with the pre-CA ($p = 0.007$; ES = 0.7) and to the post-CA ($p = 0.019$; ES = 0.61) in the AST condition was reported. Similarly, a significantly lower 30-m flying sprint time at post-48 compared with the pre-CA ($p = 0.003$; ES = 0.77) and to the post-CA ($p = 0.009$; ES = 0.68) in the COMB condition was found. Furthermore, the 30-m flying sprint time at post-CA was significantly higher in COMB than in the RST condition ($p = 0.018$; ES = 0.44) (Figure 3).

The 2-way ANOVA indicated that there was a statistically significant interaction of 50-m ($F = 5.22$; $p = 0.002$; $\eta^2 = 0.367$) sprint time. The post hoc analysis revealed a significantly lower 50-m sprint time at post-48 compared with pre-CA ($p = 0.015$; ES = 0.51) and to post-CA ($p = 0.005$; ES = 0.46) in the AST condition. Similarly, a significantly lower 50-m sprint time at post-48 hour compared with pre-CA ($p = 0.011$; ES = 0.64) and post-CA ($p = 0.001$; ES = 0.55) in the COMB condition was reported. Furthermore, the 50-m sprint time at post-48 hour was significantly lower in COMB than in the RST condition ($p = 0.017$; ES = 0.77) (Figure 4).

Biceps Femoris Stiffness

The 2-way ANOVA indicated that there was not statistically significant interaction of biceps femoris stiffness ($F = 2.235$; $p = 0.085$; $\eta^2 = 0.199$) nor the main effect of condition ($F = 1.949$; $p = 0.171$; $\eta^2 = 0.178$) and time ($F = 0.946$; $p = 0.407$; $\eta^2 = 0.095$) (Figure 5).

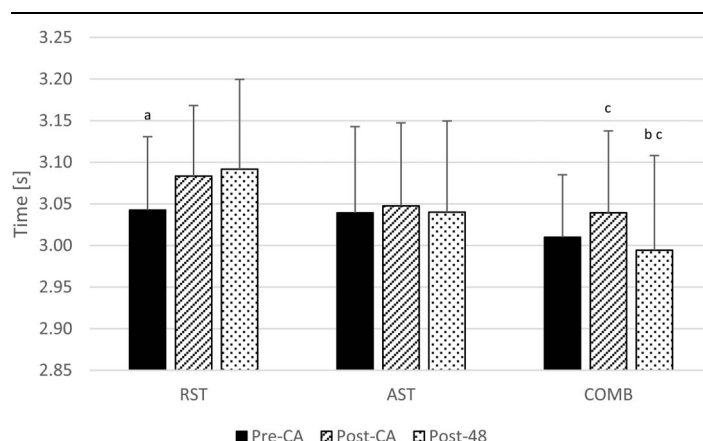


Figure 2. Preconditioning and postconditioning activity sprint results at 20 m. ^aSignificant differences compared with other time points within the condition; ^bsignificant differences in comparison to post-CA within the condition; ^csignificant differences in comparison to RST corresponding time point. RST = resisted sprint condition; AST = assisted sprint condition; COMB = combined sprint condition; CA = conditioning activity.

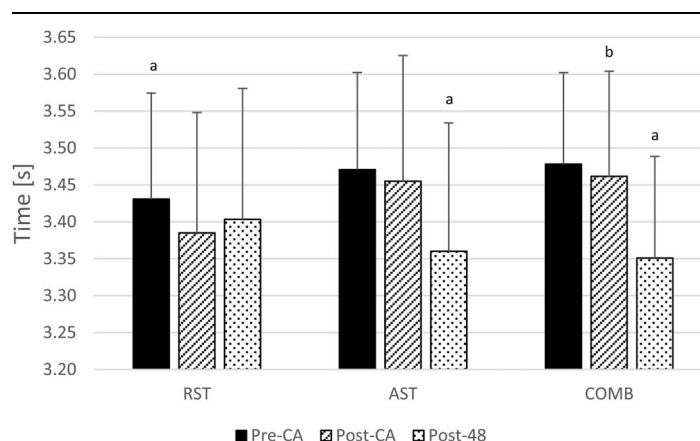


Figure 3. Preconditioning and postconditioning activity flying start sprint results at 30 m. ^aSignificant differences compared with other time points within the condition; ^bSignificant differences compared with RST corresponding time point. RST = resisted sprint condition; AST = assisted sprint condition; COMB = combined sprint condition; CA = conditioning activity.

Creatine Kinase Activity

Friedman's test showed differences in creatine kinase activity between pre-CA, post-CA, and post-48 hour (test = 14.000; $p = 0.016$; Kendall's $W = 0.280$). Pairwise comparisons demonstrated that the creatine kinase activity at post-48 during AST was significantly lower than during RST ($p = 0.028$; ES = -0.73) and COMB ($p = 0.028$; ES = 1.48) (Figure 6).

Discussion

This study aimed to compare the effectiveness of various sprinting priming protocols on PAPE and priming in sprint performance. The main finding is that the AST and COMB priming sessions improved sprint performance after 48 hours of recovery. Another outcome worth mentioning is the specificity of each condition on subsequent sprint performance. The RST contributed to the acute and delayed decline in 20-m sprint performance (by increasing time), but an improvement in the 30-m flying sprint time was noted for both time points, consequently with no effect on the 50-m sprint time.

However, AST had no acute and delayed effect on the 20-m sprint time but improved the 30-m flying sprint time 48 hours later, thus, as a result, also the 50-m sprint time. Similarly, following the COMB protocol, the effect was insignificantly better (-1.7 vs. -2.2% in the case of the 50-m sprint time). Moreover, 48 hours after the AST, the creatine kinase activity reached the lowest values compared with the other 2 exercise protocols.

One of the novel aspects of this study is the performance evaluation 48 hours after the priming session. Most studies examined the use of the 6-hour priming window (10,13,15,31,32), whereas 24-hour or 48-hour delay was less frequently analyzed (28,39). However, when the competition takes place in the morning, the use of the 6-hour priming approach seems impractical. To the best of our knowledge, this is the first study that examined the impact of sprints on priming above 6 hours and compared the impact of different types of sprints (resisted vs. assisted vs. a combination of both) in a priming session. In this study, the priming effect in 50-m sprint time occurred after AST and COMB but not after RST conditioning. This was mainly attributed to the improvements in the 30-m flying sprint time

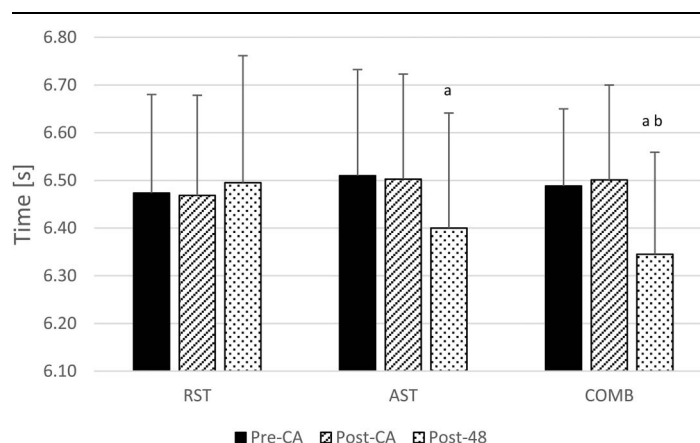


Figure 4. Preconditioning and postconditioning activity sprint results at 50 m. ^aSignificant differences compared with other time points within the condition; ^bSignificant differences compared with RST corresponding time point. RST = resisted sprint condition; AST = assisted sprint condition; COMB = combined sprint condition; CA = conditioning activity.

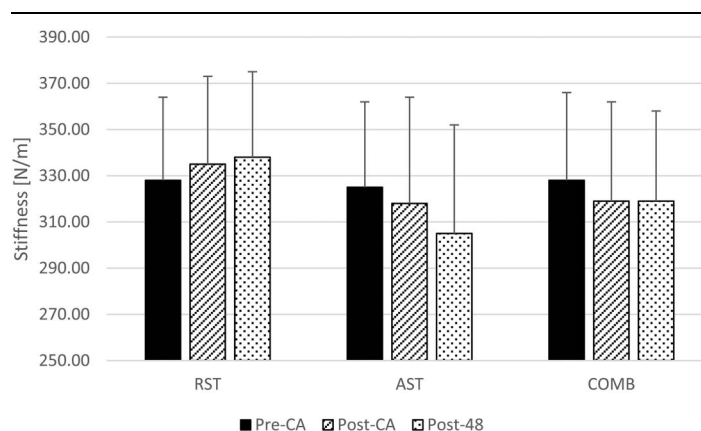


Figure 5. Changes in the stiffness of biceps femoris between preconditioning and postconditioning activity. Significant differences in comparison to RST corresponding time point. RST = resisted sprint condition; AST = assisted sprint condition; COMB = combined sprint condition; CA = conditioning activity.

because no effects were found on the 20-m sprint time 48 hours later. However, RST conditioning also significantly improved the 30-m flying sprint time but at the same time impaired the 20-m sprint time, resulting in no effect on the 50-m sprint time. Resisted sprinting are commonly used method to overload sprinting acceleration performance, whereas assisted sprinting has been proposed for maximal velocity improvements (17,30), mainly because of a velocity-specific adaptation to sprint training (21). Mentioned recommendations are related to long-term training effects, which do not directly have to mimic acute effects; however, recent studies also showed that performance enhancement using PAPE (12,22) and priming (28) may be movement and velocity specific. Consistent with this, the AST conditioning used in this study improved the 30-m flying sprint time. Nevertheless, the results after applying RST conditioning in this study, as well as findings from a recent study by Matusiński et al. (27), indicate that RST can also be used to immediately improve flying sprint time in female sprinters. Matusiński et al. (27) showed that a single resisted sprint (20 m with 10% of body mass) acutely improved the 20-m flying sprint time. In both studies, the same resistance was used (10% of body mass), which could only slightly reduce acceleration and as indicated by Petrakos et al.

(30), low-loaded resisted sprints (<10% of body mass) may be also useful in improvements in maximal sprint velocity. However, this study reported an acute and delayed increase in the 20-m sprint time, whereas Matusiński et al. (27) noted an improvement in the 20-m sprint time. The reason for these differences could be explained by the greater volume of conditioning used compared with the study by Matusiński et al. (27) (4 set vs. 1 set) and, as a result, excessive fatigue. This could partly be explained by in fact insignificant but different trend (to increase) in the stiffness of the biceps femoris after the RST condition as compared with the other conditions. Previous studies have indicated that a subtle increase in the stiffness of the muscles involved in the exercises was concomitant with performance decrease (1,35,37). In addition, the study by Higashihara et al. (18) showed that biceps femoris activity is significantly greater during the acceleration phase of sprinting compared with the maximum velocity. It is possible that higher mechanical demands placed on the biceps femoris during RST led to a slight increase in biceps femoris stiffness and creatine kinase activity resulting in acute and delayed impairment of 20-m sprint time. However, the significant improvement in 30-m flying sprint time after RST noted in this study is surprising and casts doubt on this theory.

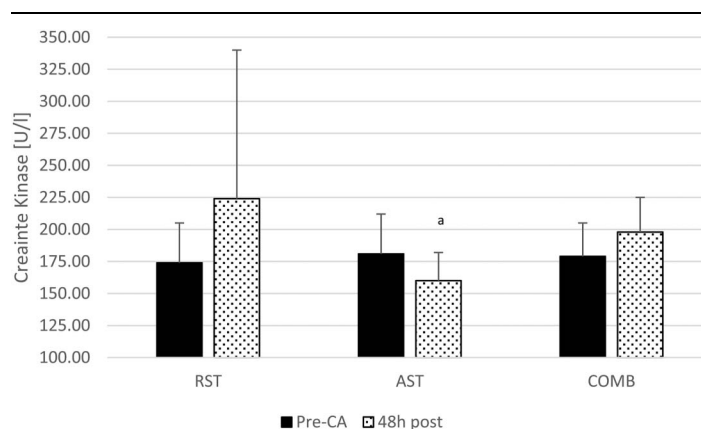


Figure 6. Changes in the creatine kinase activity between preconditioning and postconditioning activity. ^aSignificant differences compared with the corresponding time point in other conditions. RST = resisted sprint condition; AST = assisted sprint condition; COMB = combined sprint condition; CA = conditioning activity.

To the best of our knowledge, there are no available data regarding resisted and assisted conditioning during priming sessions, although only 1 directly compared in PAPE protocol among sprinters (26), which limits the possibility of comparing our results with other studies. Nevertheless, significant knowledge, training clues, and perspectives for future investigations can be derived from the current data. Matusiński et al. (26) found that 3×30 m resisted sprint conditioning with a load of 10% body mass acutely enhanced 50-m sprint time, whereas 3 sets of 40-m assisted sprints, also with 10% body mass, have no impact on 50-m sprint time performed 8 minutes later in female sprinters. The authors suggested that this might be because of excessive velocity, which caused overstriding and a loss of coordination, which did not allow for acute enhancement of sprint performance (8). Despite the lower pulling force used in this study, the AST conditioning failed to acutely improve the 50-m sprint time, by contrast, which was effective in eliciting a priming effect. In summary, our results suggested that low-loaded AST and COMB conditioning contributed to the improvement of the 50-m sprint time 48 hours after activation, which was mainly attributed to the improvements in the 30-m flying sprint time, rather than the 20-m sprint time. However, in both AST and COMB conditions, none of these protocols led to acute improvements in sprint performance at any distance. Therefore, just because the CA was ineffective in acute performance enhancement, it does not necessarily mean it will be ineffective in priming and vice versa suggesting that the PAPE and priming phenomena may be independent of each other. Furthermore, a significant reduction in creatine kinase activity post-48 was found only after AST conditioning; hence, even shorter recovery intervals might be sufficient to achieve a priming effect in such protocols.

The procedure for this study has several limitations that should be mentioned. First, only trained female sprinters participated in this study, and only 50-m sprint time was analyzed (but with 2 splits; 0–20 m and 20–50 m), which means that extrapolation of these results to other populations or distances should be done with caution. In addition, this study compared the effects of 3 different priming sessions without including the control condition and only used a single rest interval for both PAPE and priming. Moreover, we did not analyze changes in sprint mechanics.

Practical Applications

A most relevant application of this study is that combination of resisted and assisted or assisted-only sprinting using low volume (2 sets of each modality or 4 sets of assisted-only of 40 m for each) and low pulling forces (10% of body mass for resisted and ~6–7% for assisted) improved 50-m sprint time after 48 hours in female sprinters. The results also suggest that this effect was mainly attributed to the improvements in the 30-m flying sprint time with no impact on the 20-m sprint time. Therefore, when possible, coaches and athletes should consider a short sprinting bout performed 48 hours before the competition. Moreover, considering the given outcomes, coaches could try to adapt these protocols in the weekly schedule when maximum velocity is prioritized in training.

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